

```

1  #include <stdio.h>
2  int main(){
3      int n;
4      printf("Enter number of processes: ");
5      scanf("%d", &n);
6
7      int arrivalTime[10];
8      int burstTime[10];
9      int remainingTime[10];
10     int waitingTime[10] = {0};
11     int turnaroundTime[10];
12     int completed = 0;
13
14     for(int i = 0; i < n; i++)    {
15         printf("Enter arrival time of P%d: ", i + 1);
16         scanf("%d", &arrivalTime[i]);
17
18         printf("Enter burst time of P%d: ", i + 1);
19         scanf("%d", &burstTime[i]);
20
21         remainingTime[i] = burstTime[i];
22     }
23
24     int currentTime = 0;
25
26     while(completed < n)    {
27         int shortest = -1;
28         int minRemaining = 9999;
29
30         for(int i = 0; i < n; i++)
31         {
32             if(arrivalTime[i] <= currentTime &&
33                 remainingTime[i] > 0 &&
34                 remainingTime[i] < minRemaining)
35             {
36                 minRemaining = remainingTime[i];
37                 shortest = i;
38             }
39         }
40
41         if(shortest == -1)
42         {
43             currentTime++;    // CPU idle
44             continue;
45         }
46
47         remainingTime[shortest]--;
48         currentTime++;
49
50         if(remainingTime[shortest] == 0)
51         {
52             completed++;
53             turnaroundTime[shortest] =
54                 currentTime - arrivalTime[shortest];
55             waitingTime[shortest] =
56                 turnaroundTime[shortest] - burstTime[shortest];
57         }
58     }
59
60     int totalWT = 0, totalTAT = 0;
61
62     printf("\nPID\tAT\tBT\tWT\tTAT\n");
63     for(int i = 0; i < n; i++)    {
64         printf("P%d\t%d\t%d\t%d\t%d\n",
65             i+1, arrivalTime[i], burstTime[i],
66             waitingTime[i], turnaroundTime[i]);
67
68         totalWT += waitingTime[i];
69         totalTAT += turnaroundTime[i];
70     }
71
72     printf("Average WT = %.2f\n", (float)totalWT / n);
73     printf("Average TAT = %.2f\n", (float)totalTAT / n);
74 }

```